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$$f(x, y) = \sin(xy^2)$$

1ste partiële afgeleiden:

$$\begin{cases} \frac{\partial f}{\partial x}(x, y) = y^2 \cos(xy^2) \\ \frac{\partial f}{\partial y}(x, y) = 2xy \cos(xy^2) \end{cases}$$

2de partiële afgeleiden:

$$\begin{cases} \frac{\partial^2 f}{\partial x^2}(x, y) = -y^4 \sin(xy^2) \\ \frac{\partial^2 f}{\partial y^2}(x, y) = 2x \cos(xy^2) - 4x^2 y^2 \sin(xy^2) \\ \frac{\partial^2 f}{\partial y \partial x}(x, y) = 2y \cos(xy^2) - 2xy^3 \sin(xy^2) \\ \frac{\partial^2 f}{\partial x \partial y}(x, y) = 2y \cos(xy^2) - 2xy^3 \sin(xy^2) \end{cases}$$

Hessiaanse matrix:

$$D^2 f(x, y) = \begin{pmatrix} -y^4 \sin(xy^2) & 2y \cos(xy^2) - 2xy^3 \sin(xy^2) \\ 2y \cos(xy^2) - 2xy^3 \sin(xy^2) & 2x \cos(xy^2) - 4x^2 y^2 \sin(xy^2) \end{pmatrix}$$

References